

LIST OF PROBLEMS: SIPV AUCTIONS

ECO403/503–INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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1. Consider a sealed-bid first-price auction with three buyers, where the private values of the buyers are independent and uniformly distributed over $[0, 2]$. Find a symmetric equilibrium of the game and compute the seller's expected revenue.
2. Suppose there are two bidders with private values that are distributed independently according to the distribution $F(x) = x^a$ over $[0, 1]$, where $a > 0$.¹ Find the symmetric equilibrium bidding strategies in a first-price auction.
3. Suppose there are two bidders with private values that are distributed independently according to a Pareto distribution $F(x) = 1 - (x + 1)^{-2}$ over $[0, \infty)$. Show by direct computation that the expected revenues in a first- and second-price auction are the same.
4. **Selling an object at the monopoly price vs. running a first-price auction.** Nurassyl is interested in selling his mint condition collectible comic book *Amazing Fantasy No. 15*.² Assume there are n buyers and that buyer i 's private value for the comic book (in thousands of dollars) is V_i , which is uniformly distributed over $[0, 1]$. The private values of the buyers are independent.

Instead of conducting an auction, Nurassyl intends on setting a price for his comic book, publicizing this price, and selling the comic book only to buyers who are willing to pay this price; if no buyer is willing to pay the price, the comic will not be sold, and if more than one buyer is willing to pay the price, the comic will be sold to one of them based on a fair lottery that gives each of them equal probability of winning.

- (a) Find Nurassyl's expected revenue as a function of the price x that he sets. *Hint: Try first to think about the expected demand for the comic book, as a function of the price x .*
 - (b) Find the price x^m that maximizes Nurassyl's expected revenue.
 - (c) What is the maximal expected revenue that Nurassyl can obtain, as a function of the number of buyers n ?
 - (d) Compare Nurassyl's maximal revenue with the revenue he would gain if he sells the comic book by way of a sealed-bid first price auction. For which values of n does a sealed-bid first price auction yield a higher revenue?
5. Suppose that $\mathcal{V} = [0, \bar{v}]$ is a bounded interval. Show that in a symmetric sealed-bid second-price auction with independent private values, the strategy profile in which buyer 1 bids $\bar{v}$ and all other buyers bid 0 is an (asymmetric) equilibrium. Is it also an equilibrium in a sealed-bid first-price auction?
 6. Akbobek and Uldana are the only buyers in a sealed-bid first-price auction of medical devices. Akbobek knows that Uldana's private value is uniformly distributed over $[0, 2]$, and that Uldana's strategy is $\beta(v) = \frac{v^2}{3} + \frac{v}{3}$.

¹This is called a power distribution.

²Some pop culture trivia for you: This comic book marks the first appearance of Spider-Man. In 2021, it became the record-holder as the most valuable comic when it was sold for US \$3,600,000 at an auction.

- (a) What is Akbobek's optimal strategy?
 - (b) What is Akbobek's expected payment if she implements this optimal strategy (as a function of her own private value)?
7. Suppose that the private values of two buyers are independent and uniformly distributed over the set $\{0, 1, 3\}$, i.e., each buyer has three possible private values. The bids that the buyers can submit must be nonnegative integers.
- (a) Consider a sealed-bid first price auction. Find all the equilibria of such an auction under the assumptions above.
 - (b) Find all the equilibria, under the same assumptions, when the auction is a sealed-bid second-price auction.
 - (c) Compare the seller's expected revenue under both auction methods. Have you discovered something interesting?
8. Denote by R^I the seller's revenue in a sealed-bid first-price auction, and by R^{II} the seller's revenue in a sealed-bid second-price auction. Find the variance of R^I and of R^{II} when there are n buyers, whose private values are independent and uniformly distributed over $[0, 1]$. Which of the two is lesser?
9. Consider the SPA with n buyers whose private values are independent and uniformly distributed over the interval $[0, 1]$. In this exercise you will explore some of the asymmetric equilibria of this model.
- (a) Find an asymmetric equilibrium at which every buyer wins with probability $\frac{1}{n}$.
 - (b) Now for a small challenge: Can you find an equilibrium at which every buyer i wins with probability α_i , for any collection of nonnegative numbers $\alpha_1, \alpha_2, \dots, \alpha_n$ whose sum is 1?
10. Consider a symmetric FPA with independent private values with n buyers whose cumulative distribution is given by $F(v) = v^2$ over $[0, 1]$.
- (a) Find the symmetric equilibrium.
 - (b) Compute the interim expected payment of player i when her value is v_i .
 - (c) Compute the ex-ante expected payment of player i .
 - (d) Compute the seller's expected revenue π .